

WQU Capstone

Assignment: Group Formation & Topic Selection

This form **must be submitted in your online course room immediately after the course begins**. Failure to submit your group formation and topic selection will result in cancellation from the course at the start of Week 2. If you are proposing your own topic, you are required to send this form to Student Support using the Help Center chatbot prior to the M1 deadline for instructor approval.

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Name of group member(s)	Wickliff Siriga Ogari
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Select your track/topic from the dropdown menu	8. Machine Learning (Deep) Investment Strategies
<i>If response above is "Other", insert your proposed topic here</i>	

This section **is required** to provide additional information regarding your topic as this will help instructors determine if they would be a good fit to mentor your group for the duration of the project. If you opted to propose your own topic, this information will be used for approval purposes (ie. whether you have access to necessary data, etc.).

The proposed topic of interest is "Application of Machine Learning in Risk Management". Since the great financial crisis, risk management has been a key concern for financial institutions. Therefore, this topic presents us with the opportunity to take on the challenge of utilizing machine learning to mitigate financial risks including market risk, credit risk, operational risk, fraud risk. Despite the growing adoption of machine learning in risk management, most financial institutions still rely on static models which often fail to capture complexities of real time market shifts. This project addresses this gap by developing an explainable machine learning framework specifically for credit and fraud risk - where the intersection of customer behavior and transaction anomaly is critical. The machine learning framework aims to improve predictive accuracy in credit and fraud risk by leveraging advanced machine learning algorithms to process diverse, unstructured datasets(e.g social media, digital footprint logs). The projected setbacks include, limited availability of unstructured data, skill gap in processing of unstructured data, and inherent issues regarding data quality & algorithmic bias.

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Assignment: Group Formation & Topic Selection

Statement of integrity: *By typing my name in the text box below, I confirm that I am willingly entering into group formation with the students listed above and have confirmed the common interest in the topic for this project. I understand that failure to collaborate effectively with the other students of the group will result in grades of 0 and automatic isolation. If I am proposing my own topic I am required to submit this form to the Help Center chatbot prior to the course start date for instructor approval.*

Full Legal Name	Taoufiq OUEDRAOGO
Date Submitted	23/30/2026



PROBLEM STATEMENT

MScFE Capstone Project

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Group 14074

April 1, 2026

Project Track

Practical Track - Machine Learning (Deep) Investment Strategies: Quantitative Credit Risk Modeling

Problem statement

Traditional credit risk models, including Logistic Regression and standard credit scoring frameworks, assume that the relationship between borrower characteristics and probability of default (PD) is linear and stable over time (Bello 119).

However, empirical studies have shown that risk and particularly credit risk cannot be modeled linearly but rather non-linearly as a function of multiple variables, including internal borrower data, borrower behavior, macro and micro-economic conditions. Further, historical data from events like the 2008 financial crisis and the recent COVID-19 pandemic indicate that credit risk is highly non-linear and sensitive to macro and microeconomic shifts (Bello 110; Suhadolnik et al. 2, 20). Models calibrated during expansionary, low-volatility periods will systematically underestimate risk during downturns.

Drawing an analogy from asset pricing, the Capital Asset Pricing Model (CAPM) is extended by the Fama-French three-factor model by incorporating additional systematic risk factors such as size, value beyond market beta. Similarly, this project proposes to decompose credit risk into:

- Idiosyncratic borrower risk: captured by traditional credit features (leverage, interest coverage, liquidity, ...)
- Systematic macro-regime risk: driven by the macroeconomic environment and market regime at the time of assessment

This project aims to develop a macro-conditional and regime-dependent credit risk framework, where the probability of default is explicitly modeled as:

$$PD = Pr(\text{default} | X_{\text{borrower}}, M_t, R_t)$$

- X_{borrower} : borrower-specific characteristics
- M_t : macroeconomic variables at time t
- R_t : latent market regime at time t

The objective is to determine whether incorporating macroeconomic conditions and latent regimes leads to statistically and economically significant improvements in credit risk modeling.

Goals and Objectives

The goal is to determine if adding latent regime shift dynamics and macroeconomic variables using advanced machine learning considerably enhances the resilience, calibration, and prediction accuracy of probability of default (PD) models across several economic cycles.

Hypothesis: ***“Macroeconomic and market regime information significantly improves predictive accuracy of PD models”***

Objective 1: Data Pipeline

- Load and align borrower-level loan data (Lending Club dataset) with macroeconomic time series and adjust frequency to monthly.
- Microeconomic data: Federal Funds Effective Rates (FEDFUNDS), Unemployment rate (UNRATE), Consumer Price Index for All Urban Consumers: All Items (CPIAUCSL), ICE BofA US High Yield Index Option-Adjusted Spread (BAMLH0A0HYM2), and Real Gross Domestic Product (GDPC1).
- Source: Federal Reserve Bank of St. Louis (FREDAPI)
- Feature engineering: Calculation of financial ratios (leverage, interest coverage ratio), cash flow volatility, and macroeconomic time series transformations

Objective 2: Credit Risk Modeling (Base Models)

- Implement and evaluate benchmark models: Logistic Regression, Gradient Boosting (LightGBM / XGBoost)
- Input: borrower-level features only
- Estimate probability of default:

$$\text{Probability of default } PD_{\text{baseline}} = Pr(\text{default} | X_{\text{borrower}})$$

Objective 3: Macroeconomic-Conditional Extension

- Incorporate macroeconomic variables: interest rates, inflation, unemployment, credit spreads, GDP growth proxies, ...
- Extended model:

$$PD = Pr(\text{default} | X_{\text{borrower}}, M_t)$$

- Evaluate the performance between the extended models and the base models.

Objective 4: Regime Detection Layer

- Identify latent market regimes using unsupervised machine learning methods:
 - Hidden Markov Models (HMM)
 - Probabilistic clustering: Gaussian Mixture Models
 - Hierarchical clustering
 - Exclusive clustering: K-Means clustering
- Define interpretable regimes:
 - Volatility: low stress vs high stress
 - Growth: expansion vs recession
 - Liquidity: low vs high
- Assign regime label R_t to each observation

Objective 5: Regime-Dependent Credit Model

- Estimate regime-aware models:

$$PD = Pr(\text{default} | X_{\text{borrower}}, M_t, R_t)$$

- Compare:
 - Baseline vs macro vs regime-aware models
 - Stability across economic regimes

Objective 6: Model Evaluation

Evaluate all three model tiers using a structured framework:

Dimension	Metric	Purpose
Discrimination	ROC-AUC / PR-AUC	Ranking ability across borrower risk levels
Calibration	Brier Score	Probabilistic accuracy of PD estimates
Calibration	Calibration curves	Visual reliability of predicted vs. observed default rates
Robustness	Per-regime AUC	Stability of model performance across macro states
Robustness	Time-split OOS test	Out-of-sample generalization across economic cycles
Stress	2008-shock PD response	PD surface sensitivity under crisis-like macro conditions

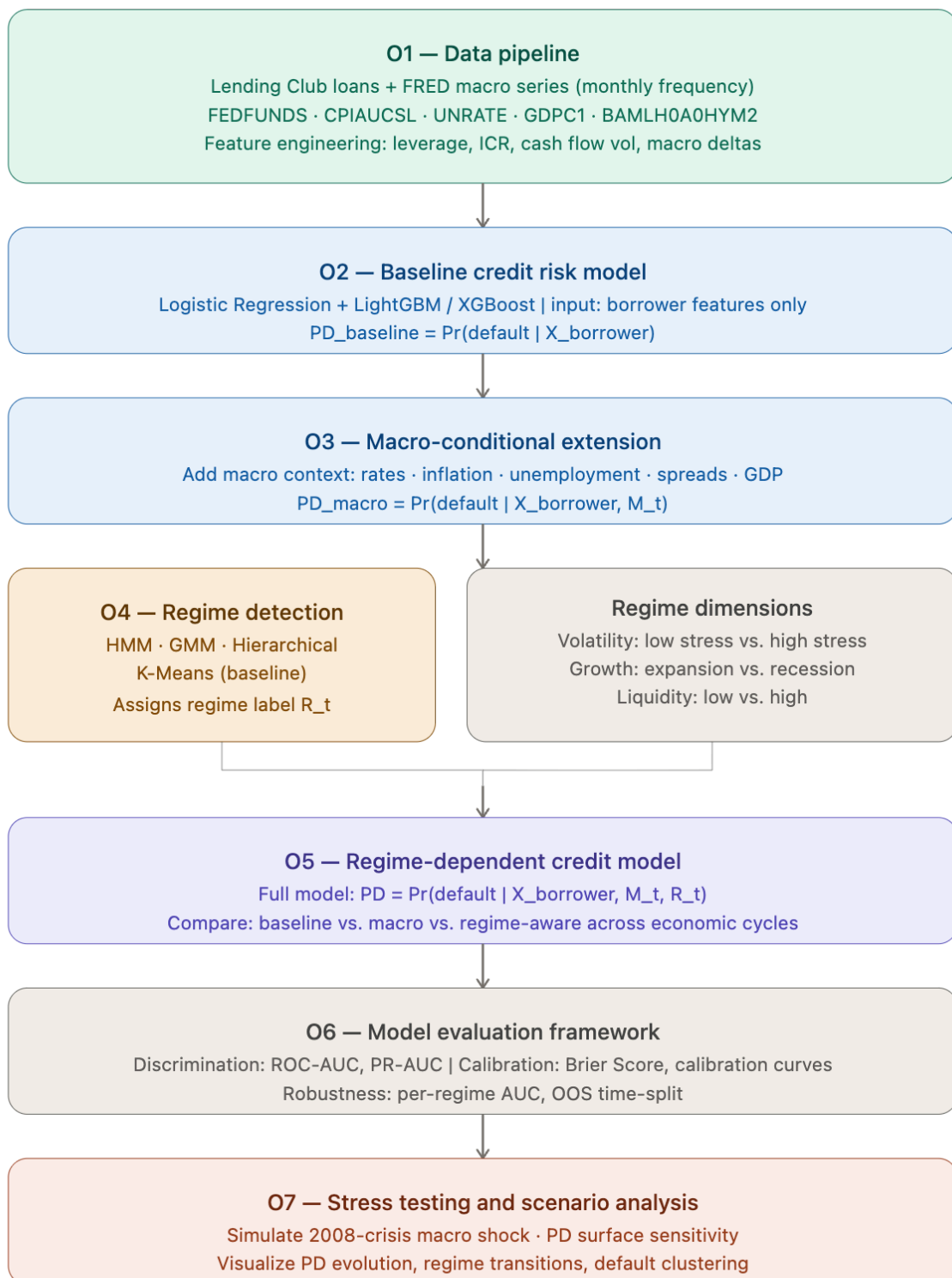
Objective 7: Stress Testing and Scenario Analysis

- Simulate adverse macroeconomic scenarios (for example, 2008 financial crisis macro conditions)
- Analyze PD sensitivity to macro shocks and default clustering effects
- Visualize PD surface evolution across macro states and regime transitions

Libraries and Data Sources

Category	Details
Data manipulation	pandas, numpy
Macro data access	fredapi (FRED API) or manual CSV download
Baseline modeling	scikit-learn, xgboost, lightgbm
Regime detection	hmmlearn, scikit-learn (GMM, KMeans, Agglomerative)
Evaluation & calibration	scikit-learn (metrics, calibration)
Visualization	matplotlib, seaborn, plotly
Loan dataset	Lending Club public dataset: Kaggle (2007–2018, labelled defaults)
Macro dataset	FRED (Federal Reserve Bank of St. Louis): free public API

Code Design Plan



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LITERATURE REVIEW AND COMPETITOR ANALYSIS

MScFE Capstone Project

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Problem Statement

Credit risk models have long relied on logistic regression and scorecard-based approaches. These methods work reasonably well under normal conditions, but they share a fundamental assumption that tends to break down in practice: that the link between a borrower's characteristics and their likelihood of defaulting stays roughly constant over time.

The 2008 financial crisis and the COVID-19 shock made this assumption hard to defend. Default rates spiked in ways that static models simply didn't anticipate, not because the borrowers looked different on paper, but because the macro environment had shifted entirely. A model trained on 2005-2007 data had no way of knowing what a 10% unemployment rate or near-zero interest rates would do to repayment behavior.

What we're trying to do in this project is account for that. Instead of treating credit risk as a fixed function of borrower features, we want to condition it on the macroeconomic environment and on which "regime" the economy is in at a given point in time whether that's a period of expansion, stress, or something in between.

The Fama-French model gave us a useful way to think about this. Just like equity returns can't be explained by market beta alone and benefit from adding size and value factors, credit risk probably can't be captured by borrower characteristics alone. There's a systematic component driven by macro conditions and market regimes that static models ignore.

So the framework we're building models PD as a function of three things: the borrower's own features, macroeconomic variables at the time of assessment, and a latent regime label estimated from the data. The core question we want to answer is whether adding those last two components actually improves prediction in a meaningful way not just on paper, but across different economic periods including stress scenarios.

$$PD = Pr(\text{default} | X_{\text{borrower}}, M_t, R_t)$$

- X_{borrower} : borrower-specific characteristics
- M_t : macroeconomic variables at time t
- R_t : latent market regime at time t

Literature Review

Introduction

The 2008 financial crisis pushed credit risk modeling to the top of the research agenda in ways it had not been before. Banks lost enormous amounts of money, and a big part of the reason was that their models were not built to handle the kind of conditions that emerged. Since then, research has moved steadily toward approaches that can handle non-linear relationships and adapt to shifting economic conditions, rather than assuming the world stays roughly the same year after year.

This review covers three areas that directly shape this project. First, the classical credit risk models that still dominate practice and where they consistently fall short. Second, the growing body of work on connecting default probability to the broader macroeconomic environment. Third, the regime-switching literature, which addresses the fact that borrower behavior and default patterns do not look the same across all economic periods.

1. Traditional Credit Risk Models and Their Limits

Most discussions of credit risk in the academic literature start with Merton (1974). His structural model frames default as the point where a company's asset value drops below what it owes – a clean piece of theory, but one that depends on inputs that cannot be directly observed in practice. That gap between theory and application pushed researchers toward reduced-form models, notably Jarrow and Turnbull (1995) and Duffie and Singleton (1999), which treat default as a probabilistic event governed by a hazard rate rather than something derived from unobservable asset dynamics. These caught on in practice but came with their own limitations: they still assume the relationship between a borrower's characteristics and their default probability does not change much over time.

In consumer lending, logistic regression has been the default tool for decades. Altman's Z-score (1968) is the most cited version of this idea, a fixed weighted combination of financial ratios used to classify firms as distressed or not. The model has been updated and extended many times, but the underlying structure has stayed the same: a linear mapping from inputs to risk, estimated once and applied going forward. Bello (2023) compared these classical methods to more recent machine learning alternatives and found logistic regression holds up well when data is limited but gets overtaken when non-linear patterns are present. Suhadolnik et al. (2023) frame the problem more bluntly: the real issue is not that the model is too simple, it is that it treats economic conditions as irrelevant, which they argue is a structural flaw rather than a tuning problem.

2. Machine Learning Applied to Credit Risk

Ensemble methods started getting serious attention in credit risk around 2015. Lessmann et al. (2015) compared 41 classifiers across multiple datasets and found that boosting methods

consistently deliver stronger AUC scores than classical alternatives. They were careful to flag something important though strong AUC does not guarantee well-calibrated probabilities, and in credit risk that distinction matters. A lender needs PD estimates that reflect actual default rates, not just a model that ranks borrowers correctly relative to each other. Brown and Mues (2012) made a complementary point about gradient boosting specifically: it handles missing values and feature interactions better than logistic regression and does not require the extensive manual feature engineering that traditional scorecards demand.

One pattern worth noting across this literature is how often the time dimension gets ignored in model evaluation. Random train-test splits are still common practice in published studies, which can produce results that look good on paper but would not hold up in deployment. A model trained on data from 2015 and evaluated on data from 2010 has effectively been given information it could not have had at the time. This project avoids that by using a strict time-based split throughout, with evaluation only on loans issued after a fixed cutoff date.

3. Macroeconomic Conditioning in Credit Models

Connecting credit risk to macroeconomic conditions is not a new idea, but embedding it properly into a machine learning pipeline is still relatively underdeveloped. Wilson's CreditPortfolioView (1997) was one of the earlier frameworks to do this systematically, modeling portfolio-level default rates as a function of GDP growth, unemployment, and interest rates through a system of equations. It showed that macro variables carry real explanatory power for default behavior at the aggregate level, even if firm-level characteristics dominate at the individual borrower level.

The regulatory framework reflects this too. Basel II and III draw a formal distinction between through-the-cycle and point-in-time PD estimates (BCBS, 2006). Through-the-cycle estimates try to average out cyclical variation; point-in-time estimates reflect where risk stands given current economic conditions. A model that conditions on macro variables is essentially doing the latter; it is trying to capture how risk shifts as the economy moves through different phases, rather than treating the cycle as noise to be smoothed away.

The choice of macro variables in this project is grounded in empirical findings. Figlewski et al. (2012) found that unemployment and industrial production growth have statistically significant effects on corporate default rates, holding firm-level factors constant. Blanco et al. (2020) showed that high-yield credit spreads including the BAMLH0A0HYM2 series used here carry forward-looking information about aggregate default risk that borrower data alone does not reflect. Both findings point toward the same conclusion: leaving macro variables out of a credit model means leaving out information that genuinely moves default rates.

4. Regime-Switching Frameworks

Hamilton (1989) introduced Hidden Markov Models to economics through a deceptively simple observation: US GDP growth does not behave as if it were drawn from a single stable distribution. It behaves more like a process that switches between two states expansion and contraction with the

active state hidden from direct observation but inferable from the data. That idea opened up a whole line of research into latent regime models for financial and economic time series.

The implications for credit risk are significant. Nickell et al. (2000) showed that credit rating transition probabilities vary substantially across economic states. A borrower sitting at a given rating is meaningfully more likely to be downgraded during a recession than during an expansion, even if their underlying financials look similar. A model that assumes these probabilities are constant is going to get things systematically wrong every time the economic environment shifts.

Bai et al. (2019) brought this logic into a machine learning context by training separate gradient boosting models for each regime identified by a two-state HMM. Their regime-specific models outperformed a single pooled model on both AUC and Brier score, which is probably the most direct empirical precedent for what this project does. The approach here extends theirs in a few ways: three regimes instead of two, multiple clustering methods compared rather than HMM alone, and the regime label included as a feature in the model rather than used to partition the training data entirely which keeps the model from losing information during regime transitions.

Gaussian Mixture Models are also included as an alternative to HMMs. Frühwirth-Schnatter (2006) makes the case that GMMs can be more appropriate when there is no strong reason to assume regime transitions are Markovian. Whether that assumption holds for the specific macro feature set used here is an open empirical question, and comparing the two methods directly is part of how this project tries to answer it.

Competitor Analysis

In the context of this capstone, “competitors” refer to other academic research, commercial and regulatory systems that have tried to solve a similar problem i.e. how to estimate default probability accurately and robustly across economic conditions.

The goal is to highlight where current methods perform well, where they fall short, and how this project can add value.

Comparison Table

We use the SWOT framework to do our analysis

1. Commercial platform

S&P Global Credit Analytics

Strengths • Covers a huge range of corporate and structured credit products in one platform • PD, LGD, and EAD all integrated together • Long history of regulatory acceptance by banks and supervisors	Weaknesses • Fully closed • No access to the underlying models, no way to replicate results • Built for institutional credit, consumer loans are not really the focus • Way too expensive for academic or research use
Opportunities • Our project targets public consumer loan data, a segment they essentially ignore • Being open and transparent is itself a real advantage here	Threats • Their data scale and regulatory credibility are very hard to compete with at the institutional level • Big banks are unlikely to switch away from a validated commercial product

MSCI BarraOne

Strengths • Strong macro factor integration across asset classes • Widely adopted by asset managers for portfolio-level stress testing • Sophisticated scenario analysis tools already built in	Weaknesses • Designed for portfolio risk management, not loan-level PD estimation • Regime detection is not an explicit feature • No open benchmarking or replication is possible
Opportunities • Our project addresses the loan-level borrower scoring problem that BarraOne was simply not built for	Threats • If banks start using BarraOne for macro stress scenarios, they may feel less need for academic alternatives

Moody's Analytics CreditEdge

Strengths • Validated against decades of actual default data • The framework does incorporate some macro conditioning • Strong credibility with regulators and large financial institutions	Weaknesses • Methodology is a black box • No transparency or reproducibility • Only covers corporate and institutional credit, consumer lending is largely out of scope • Cost makes it inaccessible for research
Opportunities • A transparent, open-source approach fills a real gap that CreditEdge does not cover • Consumer credit on public data is a completely different use case	Threats • Moody's has proprietary data at a scale no academic project can match • Regulatory endorsement gives them strong institutional lock-in

2. Regulatory framework

Basel IRB Framework

Strengths • The regulatory standard • Banks are required to use it under Basel II/III • Explicitly requires through-the-cycle PD calibration, which is closely related to what we are doing	Weaknesses • In practice, most banks implement it as basic logistic regression • PD estimates are smoothed and conservative by design, not dynamic • Not publicly available for benchmarking or comparison
Opportunities • Our regime-conditioning approach maps directly onto the through-the-cycle vs. point-in-time debate • ML can capture non-linearities that IRB misses	Threats • Regulators prefer interpretable, auditable models • ML faces a structural adoption barrier here • IRB has been embedded in bank processes for decades, academic proposals cannot easily replace that

3. Research papers

Gramifar, Kabir & Khan: Bayesian Networks + ML for credit risk

Strengths • Combines probabilistic graphical models with ML classifiers in an interesting way • Handles class imbalance reasonably well • Good performance on precision and recall in their test settings	Weaknesses • Their Bayesian network is static • No macro variables, no regime dynamics • Tested mainly on cross-sectional data with no temporal structure • No out-of-sample stress testing
Opportunities • Our framework adds exactly what they are missing: time-varying macro context and regime labels	Threats • As Bayesian methods become popular for interpretability, follow-up work could add macro features before we publish

Suhadolnik, Ueyama & Da Silva: ML for Enhanced Credit Risk

Strengths • Solid empirical evidence that ML outperforms logistic regression on credit data • Careful evaluation methodology • Open access and peer-reviewed	Weaknesses • Treats credit risk as a static problem • No regime detection, no macro conditioning • Train-test split does not appear to respect time ordering • Evaluation stops at AUC, no calibration or stress analysis
Opportunities • Our work is a direct extension of theirs • We test what happens when you layer macro variables and regime labels on top of their gradient boosting baseline	Threats • An updated version of their paper incorporating macro variables could close gap we are trying to fill pretty quickly

Adedoyin: Strategic Applications of AI/ML in US SMEs Risk Management

Strengths • Useful broad survey of ML adoption in SME risk management • Shows real-world demand for dynamic, data-driven credit frameworks • Good context for why this problem matters in practice	Weaknesses • No actual credit scoring model is built or tested • Findings are qualitative • No regime detection, no macro conditioning, no quantitative evaluation at all
Opportunities • Confirms there is real demand for the kind of framework we are building • SME lending is a natural next application of our consumer credit work	Threats • Risk of being cited as a technical reference when it is really just a market overview • More rigorous quantitative work needs to be visible alongside it

Our Unique Contribution

Looking at different approaches reviewed earlier, it becomes clear that most models are incomplete. Each one solves part of the problem, but none really combine everything together.

- Regulatory models are simple / reliable, but are mostly linear and do not react well to economic changes

- Commercial platforms include macroeconomic information, but often added on top rather than built directly into the model. They are also closed
- Academic machine learning models improve prediction, but they usually ignore macro conditions and regime changes

Our project tries to bring these pieces together in a single framework by:

- including macroeconomic variables directly in the model (not as a separate step)
- identifying market regimes using data-driven methods like clustering or HMM
- allowing the model to adjust depending on the current economic regime
- evaluating performance not just overall, but across different market conditions
- adding a simple stress testing setup to see how the model behaves in crisis scenarios

Below, the SWOT matrix of our project

Strengths • Combines micro (borrower) and macro information in one model • Captures non-linear relationships using ML • Includes regime detection, which many models ignore • Fully reproducible using public data	Weaknesses • More complex than traditional approaches • Needs careful data preparation and alignment • Results may depend on how regimes are defined • Smaller dataset compared to commercial systems
Opportunities • Addresses gap between academic and industry • Can be extended to other credit segments • Fits well with growing interest in macro-driven risk modeling • Provides benchmark for future open-source research	Threats • Commercial platforms have more data and resources • Similar research could emerge quickly • Regulatory constraints may limit adoption • Competition from other ML-based credit models

GitHub and Code Repository

All project code is hosted publicly at:

<https://github.com/Taoufiq-Ouedraogo/Quantitative-Macro-Credit-Risk-Modeling>

The repository has been initialized and structured in advance of the implementation work, which begins in Module 4. At this stage, the folder architecture is in place and reflects the project objectives directly.

We organized as follows:

- *data/* : scripts for gathering Lending Club loan dataset with FRED macroeconomic series
- *models/* : implementations of our 3 model: borrower-only baseline + macro-conditional extension + full regime-dependent model
- *regimes/* : unsupervised regime detection using HMM, GMM, K-Means and hierarchical clustering, along with logic for assigning regime label to each observation
- *evaluation/* : structured evaluation scripts covering ROC-AUC, PR-AUC, Brier score, calibration curve and the time-based out-of-sample split
- *stress/* : 2008-crisis macro shock simulation + PD surface visualization across macro states and regime transitions
- *notebooks/* : exploratory data analysis and interim results as they become available

Code development, initial results will be reported starting from Module 4. The repo will be updated continuously from that point forward, and access will be shared with the instructor as required.

The screenshot displays a GitHub repository page. At the top, a table lists the repository's structure:

File/Folder	Description	Last Commit
data	initialize repo with core structure	now
evaluation	initialize repo with core structure	now
models	initialize repo with core structure	now
notebooks	initialize repo with core structure	now
regimes	initialize repo with core structure	now
stress	initialize repo with core structure	now
wqu - module milestones	initialize repo with core structure	now
.gitignore	initialize repo with core structure	now
LICENSE	Initial commit	2 hours ago
README.md	initialize repo with core structure	now

Below the table, the 'README' file is expanded, showing the project title 'Quantitative-Macro-Credit-Risk-Modeling' and a description: 'This project explores how incorporating macroeconomic variables and market regimes can improve the prediction of Probability of Default (PD) in credit risk models. Traditional credit scoring models rely mainly on borrower-specific features and assume stable relationships over time. However, real-world evidence shows that credit risk is highly sensitive to economic conditions and behaves differently across market regimes. This project builds a macro-conditional and regime-aware machine learning framework to better capture these dynamics.'

On the right side of the repository page, there are sections for 'Releases' (No releases published), 'Packages' (No packages published), and 'Contributors' (1 contributor: Taoufiq-Ouedraogo).

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Issuance or underwriting of loans, also known as credit to individuals or businesses, is part of, if not one of the major businesses of financial institutions. Issuance of these loans is not without risks, chief among them is defaults – when the loan counterpart is unable to honor their loan obligations (cite). Therefore, financial institutions are always looking for better approaches of cubbing their exposure to loan defaults and minimize

the cost of these 'bad loans' which might significantly hurt their profits and ultimately their operations. Over the years, financial institutions have come to appreciate the benefits of prescreening of borrowers, which allows them to have a better understand their risk profile of the borrowers and thus helps them to make a better judgment in loan allocation and approval (cite). However, most of these prescreening methods - including qualitative framework, Linear discriminant model, Logistic Regression model- are traditional i.e., static in their app

Systematic Credit Risk Modeling under Macroeconomic Regimes

MScFE Capstone Project - Group 14074

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Abstract

Traditional credit risk models assume a stable and static relationship between borrower characteristics and the probability of default (PD). However, evidence from systemic shocks such as the 2008 financial crisis and the COVID-19 pandemic shows that this assumption breaks down under shifting macroeconomic conditions, where risk dynamics evolve across distinct regimes. This paper develops and evaluates a three-tier macro-conditional and regime-dependent credit risk framework: (1) a borrower-only baseline, (2) a macro-augmented specification, and (3) a full regime-dependent model incorporating latent macroeconomic states derived from unsupervised clustering methods, including K-Means, Gaussian Mixture Models, and Agglomerative Hierarchical Clustering. Four classification models-Logistic Regression, Decision Tree, XGBoost, and Random Forest-are evaluated across all feature frameworks using ROC AUC, Average Precision, and Brier Score. Model assessment is complemented by walk-forward backtesting and macroeconomic stress testing to ensure temporal robustness and behavior under adverse economic scenarios. Empirical results show that the regime-dependent Logistic Regression model achieves the best overall performance, reaching a ROC-AUC of 0.754 while also delivering superior calibration and interpretability. In contrast, tree-based ensemble methods exhibit limited sensitivity to feature framework augmentation, indicating that they already capture nonlinear interactions between borrower and macroeconomic variables internally. The findings suggest that compressing macroeconomic dynamics into structured regime representations provides the greatest benefit to linear models, enhancing both predictive accuracy and practical usability in credit risk applications.

Keywords: *Credit Risk, Machine Learning, Logistic Regression, XGBoost, Regime Detection, KMeans, GMM, Agglomerative Clustering, Macroeconomic Conditioning, Probability of Default, Walk-Forward Backtesting, Stress Testing.*

1. Introduction

Credit risk models have long relied on logistic regression and scorecard-based approaches. These methods work reasonably well under normal conditions, but they share a fundamental assumption that tends to break down in practice: that the link between a borrower's characteristics and their likelihood of defaulting stays roughly constant over

time. The 2008 financial crisis and the COVID-19 shock made this assumption hard to defend. Default rates spiked in ways that static models simply didn't anticipate, not because the borrowers looked different on paper, but because the macro environment had shifted entirely. A model trained on 2005-2007 data had no way of knowing what a 10% unemployment rate or near-zero interest rates would do to repayment behavior.

This project investigates the effect of the shift in macroeconomic conditions on loan repayment behavior. Instead of treating credit risk as a fixed function of borrower features, we want to condition it on the macroeconomic environment and on which "regime" the economy is in at a given point in time, whether that's a period of expansion, stress, or something in between.

The Fama-French model gave us a useful way to think about this. Just like equity returns can't be explained by market beta alone and benefit from adding size and value factors, credit risk probably can't be captured by borrower characteristics alone. There's a systematic component driven by macro conditions and market regimes that static models ignore.

This research uses models that treat probability of default (PD) as a function of: the borrower's own features, macroeconomic variables at the time of assessment, and a latent regime label estimated from the data. The core question we want to answer is whether adding those last two components actually improves prediction in a meaningful way, not just on paper, but across different economic periods, including stress scenarios.

2. Research Questions and Objectives

This project follows the following key research questions:

1. Does incorporating macroeconomic conditions improve PD model predictive performance relative to a borrower-only baseline?
2. Does incorporating latent economic regimes further improve model performance, especially during economic turbulence?
3. Which regime detection method (KMeans, GMM, AHC) best captures meaningful latent economic states?
4. Do regime-aware models improve calibration and stability of PD estimates across different economic cycles?
5. Does the proposed framework generalize well to walk-forward backtesting and macro stress scenarios?

The project aims to address these research questions by executing the following objectives:

1. Integrating and synchronizing multiple data sources, including the integration of the Lending Club loan dataset with FRED macroeconomic data series (e.g., Unemployment Rate). We achieve this using a point-in-time join method, ensuring no data leakage.

2. Implementing and comparing the Hidden Markov Model, Gaussian Mixture Models, and K-means clustering, and detecting the most statistically robust economic regimes.
3. Iteratively developing a three-tier modelling hierarchy using the Gradient Boosting (XGBoost) algorithm:

Model 1: Borrower's characteristics only (Traditional Model)

Model 2: Borrower's characteristics + macroeconomic features

Model 3: Borrower + Macro + Regime-label features

4. Comprehensively assessing the performance of the three models using AUC and calibration curves to determine if the regime-awareness of a model decreases probability drift during periods of economic shift.
5. Performing stress testing simulation of extreme economic shocks such as the 2008 economic crisis and Covid-19, and evaluating the stability of probability of default estimates, and generalizing the results across different market conditions.

3. Literature Review

The 2008 financial crisis pushed credit risk modeling to the top of the research agenda in ways it had not been before. Banks lost enormous amounts of money, and a big part of the reason was that their models were not built to handle the kind of conditions that emerged. Research has moved steadily toward approaches that can handle non-linear relationships and adapt to shifting economic conditions, rather than assuming the world stays roughly the same year after year.

This review covers three areas that directly shape this project. First, the classical credit risk models that still dominate practice, and where they consistently fall short. Second, the growing body of work on connecting default probability to the broader macroeconomic environment. Third, the regime-switching literature, which addresses the fact that borrower behavior and default patterns do not look the same across all economic periods.

3.1 Traditional Credit Risk Models and Their Limits

Most discussions of credit risk in the academic literature start with Merton (1974). His structural model frames default as the point where a company's asset value drops below what it owes, a clean piece of theory, but one that depends on inputs that cannot be directly observed in practice. That gap between theory and application pushed researchers toward reduced-form models, notably Jarrow and Turnbull (1995) and Duffie and Singleton (1999), which treat default as a probabilistic event governed by a hazard rate rather than something derived from unobservable asset dynamics. These caught on in practice but came with their own limitations: they still assume the relationship between a borrower's characteristics and their default probability does not change much over time.

The logistic regression model, which predicts the probability of a binary outcome (i.e., yes/no, 0/1, etc), has been the default tool in consumer lending for decades. Altman's

Z-score (1968) is the most cited version of this idea, a fixed weighted combination of financial ratios used to classify firms as distressed or not. The model has been updated and extended many times, but the underlying structure has stayed the same: a linear mapping from inputs to risk, estimated once and applied going forward. Bello (2023) compared these classical methods to more recent machine learning alternatives and found that logistic regression holds up well when data is limited, but gets overtaken when non-linear patterns are present. Suhadolnik et al. (2023) frame the problem more bluntly: the real issue is not that the model is too simple, but that it treats economic conditions as irrelevant, which they argue is a structural flaw rather than a tuning problem.

3.2 Machine Learning Applied to Credit Risk

Ensemble methods started getting serious attention in credit risk around 2015. Ensemble methods are machine learning techniques that combine several individual models (known as base models or learners) via boosting, bagging, or stacking to produce a single, more accurate and robust model. Lessmann et al. (2015) compared 41 classifiers across multiple datasets and found that boosting methods consistently deliver stronger Area Under the Receiver Operating Characteristic Curve (ROC-AUC) scores than classical alternatives. They were careful to flag something important, that is, a strong ROC-AUC does not guarantee well-calibrated probabilities, and in credit risk, that distinction matters. A lender needs PD estimates that reflect actual default rates, not just a model that ranks borrowers correctly relative to each other. Brown and Mues (2012) made a complementary point about gradient boosting specifically: it handles missing values and feature interactions better than logistic regression and does not require the extensive manual feature engineering that traditional scorecards demand.

One pattern worth noting across this literature is how often the time dimension gets ignored in model evaluation. Random train-test splits are still common practice in published studies, which can produce results that look good on paper but would not hold up in deployment. A model trained on data from 2015 and evaluated on data from 2010 has effectively been given information it could not have had at the time. This project avoids that by using a strict time-based split throughout, with evaluation only on loans issued after a fixed cutoff date.

3.3 Macroeconomic Conditioning in Credit Models

Connecting credit risk to macroeconomic conditions is not a new idea, but embedding it properly into a machine learning pipeline is still relatively underdeveloped. Wilson's CreditPortfolioView (1997) was one of the earlier frameworks to do this systematically, modeling portfolio-level default rates as a function of GDP growth, unemployment, and interest rates through a system of equations. It showed that macro variables carry real explanatory power for default behavior at the aggregate level, even if firm-level characteristics dominate at the individual borrower level.

The regulatory framework reflects this too. Basel II and III draw a formal distinction between through-the-cycle and point-in-time PD estimates (BCBS, 2006).

Through-the-cycle estimates try to average out cyclical variation; point-in-time estimates reflect where risk stands given current economic conditions. A model that conditions on macro variables is essentially doing the latter; it is trying to capture how risk shifts as the economy moves through different phases, rather than treating the cycle as noise to be smoothed away.

The choice of macro variables in this project is grounded in empirical findings. Figlewski et al. (2012) found that unemployment and industrial production growth have statistically significant effects on corporate default rates, holding firm-level factors constant. Blanco et al. (2020) showed that high-yield credit spreads, including the BAMLH0A0HYM2 series, carry forward-looking information about aggregate default risk that borrower data alone does not reflect. Both findings point toward the same conclusion: leaving macro variables out of a credit model means leaving out information that genuinely moves default rates.

3.4 Regime Detection and Switching Framework

Hamilton (1989) introduced Hidden Markov Models to economics through a deceptively simple observation: US GDP growth does not behave as if it were drawn from a single stable distribution. It behaves more like a process that switches between two states, expansion and contraction, with the active state hidden from direct observation but inferable from the data. That idea opened up a whole line of research into latent regime models for financial and economic time series.

The implications for credit risk are significant. Nickell et al. (2000) showed that credit rating transition probabilities vary substantially across economic states. A borrower sitting at a given rating is meaningfully more likely to be downgraded during a recession than during an expansion, even if their underlying financials look similar. A model that assumes these probabilities are constant is going to get things systematically wrong every time the economic environment shifts.

Bai et al. (2019) brought this logic into a machine learning context by training separate gradient boosting models for each regime identified by a two-state Hidden Markov Model (HMM). Their regime-specific models outperformed a single pooled model on both ROC-AUC and Brier score, which is probably the most direct empirical precedent for what this project does. The approach here extends theirs in a few ways: three regimes instead of two, multiple clustering methods compared rather than HMM alone, and the regime label included as a feature in the model rather than used to partition the training data entirely, which keeps the model from losing information during regime transitions.

Gaussian Mixture Models are also included as an alternative to HMMs. Frühwirth-Schnatter (2006) makes the case that GMMs can be more appropriate when there is no strong reason to assume regime transitions are Markovian. Whether that assumption holds for the specific macro feature set used here is an open empirical question, and comparing the two methods directly is part of how this project tries to answer it.

4. Methodology and Approach

We propose a regime-dependent credit risk model that incorporates borrower features, macroeconomic conditions at a given time t , and latent market regimes to evaluate the creditworthiness of a borrower by assessing the probability of default of the borrower. We implemented a multi-step quantitative approach, moving from synthesis of raw data to unsupervised regime detection (via Hidden Markov Model), and finally to supervised predictive modeling (via Extreme Gradient Boosting). The guiding equation is:

$$\text{Probability of Default (PD)} = P_r(\text{Default} | X_{\text{borrower}}, M_t, R_t) \dots\dots\dots(1)$$

Where:

- X_{borrower} : Borrower-specific characteristics
- M_t : Macroeconomic variables at time t
- R_t : Latent market regime at time t

4.1 Data Pipeline

The dataset combines two sources.

Borrower-level data was obtained from the Lending Club public dataset on Kaggle (500,000 rows, 2016-2018), including loan characteristics (loan amount, term, interest rate, installment, and grade), borrower profile variables (employment length, home ownership, annual income, and debt-to-income ratio), and credit history indicators (delinquencies, public records, revolving balance, and credit age). The binary target variable distinguishes Charged Off loans (default = 1) from Fully Paid loans (default = 0), with defaults representing 15.2% of completed loans.

Macroeconomic data was retrieved from the Federal Reserve Economic Data (FRED) database, including the Federal Funds Effective Rate, Unemployment Rate, Consumer Price Index, Industrial Production Index, Real GDP, BAA-10Y Corporate Credit Spread, 10-Year Treasury Yield, and 2-Year Treasury Yield. All eight series were forward-filled to handle missing dates and standardized before merging with the loan-level data.

The loan dataset was prepared through a structured preprocessing pipeline that included ordinal encoding for loan grade and sub-grade, frequency encoding for state information, logarithmic transformation and scaling of skewed continuous variables, creation of binary indicators for delinquency history, and categorization of loan amounts into seven credit bands. Macroeconomic variables were merged using the monthly loan issue date through a left join, ensuring a point-in-time design that avoids future information leakage.

To integrate borrower-level and macroeconomic information, a point-in-time join strategy was implemented by merging macroeconomic indicators with loan records using the monthly loan issuance date. This design ensures that the model only uses information available at the time of credit evaluation, thereby preventing data leakage. The merged dataset can be formally represented as:

$$D_{merged} = \{ (x_{i,t}, m_{\tau}) \mid \tau = \max(t_{macro} \leq t_{loan}) \} \dots\dots\dots (2a)$$

Where:

- $x_{i,t}$: features for the borrower i at the loan issuance time t
- m_{τ} : microeconomic indicator at time τ
- t_{loan} : time stamp of credit decision

All the macroeconomic variables were lagged by one period to prevent look-ahead bias. Macroeconomic information available at the time $t - 1$ is used for each loan issued at a time t :

$$X_t^{macro} = X_{t-1}^{macro} \dots\dots\dots (2b)$$

The lag structure (equation 2b) ensures that the model does not include information that would not be observable at the time of credit scoring.

4.2 Three-Tier Modeling Framework

Three feature spaces define the modeling hierarchy adopted in this study:

- **Framework 1 (Baseline):** Borrower-level features only

$$PD = f(X_{borrower})$$

- **Framework 2 (Macro-Augmented):** Borrower-level and macroeconomic variables

$$PD = f(X_{borrower}, M_t)$$

- **Framework 3 (Regime-Dependent):** Borrower-level variables, macroeconomic indicators, and latent economic regime labels

$$PD = f(X_{borrower}, M_t, R_t)$$

Where:

- PD : denotes the probability of default
- $X_{borrower}$: represents borrower-specific characteristics
- M_t : corresponds to macroeconomic variables observed at time t
- R_t : denotes the latent economic regime inferred from macroeconomic conditions

The hierarchical structure enables a progressive evaluation of predictive performance as additional layers of economic information are introduced into the credit risk modeling framework.

All frameworks employ a chronological train-test split based on the 80th percentile of the loan issuance date distribution, with the split occurring on July 1, 2018. This time-based partition preserves the temporal ordering of observations and prevents future information from influencing historical predictions. The training dataset contains 41,786 loans, while the test dataset includes 8,075 loans.

Four classification algorithms are evaluated under each framework:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

To address class imbalance and reduce false positive default classifications, a fixed decision threshold of 0.7 is applied across all model evaluations. This threshold selection prioritizes predictive precision while maintaining robustness in the identification of high-risk borrowers.

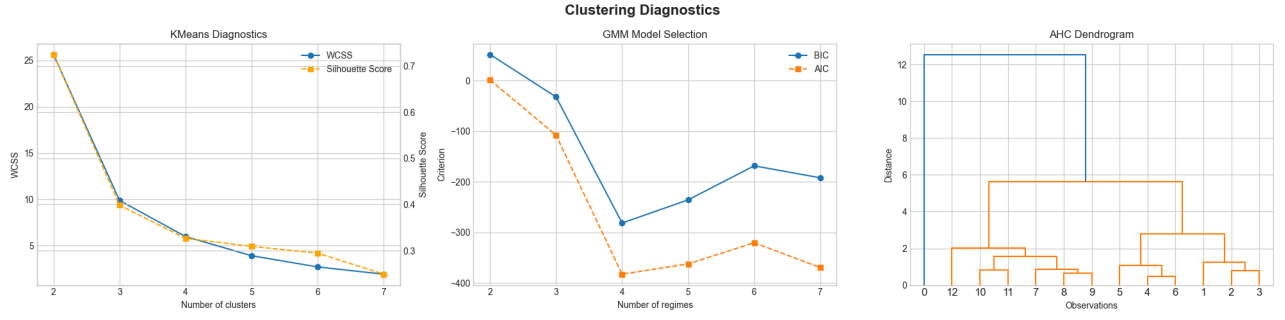
4.3 Regime Detection

Regime identification is performed using three complementary clustering approaches applied to eight standardized macroeconomic variables over a 13-month observation window. The objective is to uncover latent economic states that evolve over time and influence credit risk dynamics.

K-Means clustering is first applied with $K=3$. The optimal number of clusters is selected using the elbow criterion, where the within-cluster sum of squares (WCSS) exhibits a clear inflection at $K=3$. In addition, silhouette scores deteriorate for $K>3$, reinforcing the stability of the three-cluster solution.

Gaussian Mixture Models (GMM) with full covariance matrices are then estimated as a probabilistic alternative. Model selection using both Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) indicates a minimum around $K=4$. However, $K=3$ is retained as the preferred specification due to improved interpretability and consistency with the K-Means solution.

Hierarchical Agglomerative Clustering using Ward linkage is also performed. The dendrogram structure supports a natural partition into 2-3 dominant macroeconomic groups, aligning with the cluster structure identified by the other methods.



Visualizing Clustering Methods Selection

To further assess the dimensionality of the macroeconomic feature space, Principal Component Analysis (PCA) is conducted. The scree plot indicates that the first principal component explains approximately 80% of the total variance, suggesting that macroeconomic conditions over the sample period are largely driven by a single dominant factor.

Across all methods, the resulting regime assignments show strong consistency. A majority-vote ensemble strategy is therefore used to construct a unified regime indicator, denoted as R_{final} , which assigns each time period to its most frequently predicted cluster across methods.

Three stable macroeconomic regimes are identified:

- Regime 1: Low interest rates with elevated credit spreads, primarily observed in early 2016
- Regime 2: Rising interest rates under normalizing economic conditions, characteristic of early to mid-2018
- Regime 0: Peak rate tightening environment, observed in late 2018

The resulting R_{final} regime label is appended to Framework 2, thereby forming Framework 3, which incorporates borrower-level features, macroeconomic variables, and latent economic state information within a unified predictive structure.

4.4 Evaluation Framework

Model performance is assessed using three primary metrics that capture complementary aspects of predictive quality:

- ROC AUC, measuring ranking discrimination ability between default and non-default cases
- Average Precision (AP), capturing performance under class imbalance and emphasizing rare-event detection
- Brier Score, evaluating the calibration quality of predicted probabilities

Formally, the Brier Score is defined as:

$$\frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$$

Where p_i is the predicted probability of default and $y_i \in \{0, 1\}$ is the observed outcome.

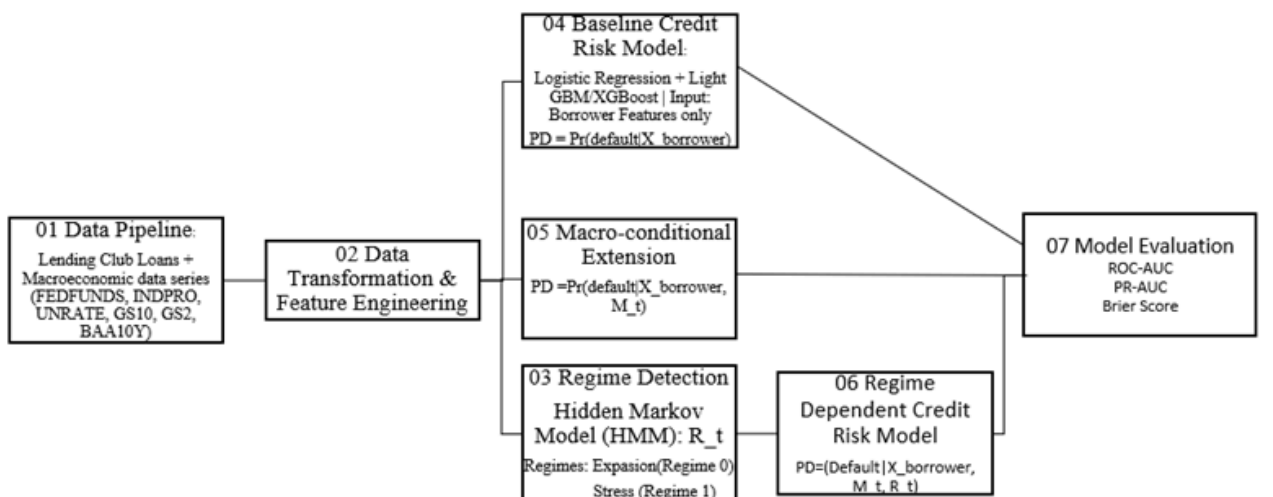
In addition to scalar performance metrics, a set of diagnostic tools is used to evaluate model behavior:

- ROC curves and calibration plots for visual assessment of discrimination and probability reliability
- Confusion matrices computed at a fixed decision threshold of 0.7 to reflect operational decision rules
- Learning curves to assess bias-variance trade-offs and sample efficiency
- Feature importance analysis to identify key drivers of default prediction across frameworks

Temporal generalization is evaluated through walk-forward backtesting using a TimeSeriesSplit procedure with five folds. In each fold, models are trained exclusively on past observations and tested on strictly subsequent data, ensuring a causally consistent evaluation and eliminating look-ahead bias.

Frameworks 1 through 3 are evaluated using previously fitted model instances, allowing direct assessment of temporal stability and robustness of each feature specification under realistic time-evolving conditions.

Figure 1 below provides a visual of the workflow of our project from developing data pipeline to stress testing and scenario analysis.



4.5 Stress Testing and Sensitivity Analysis

Macro stress testing is conducted by applying simultaneous shocks to all eight standardized macroeconomic variables within the test set. A set of ten predefined stress scenarios is considered to capture a broad spectrum of economic conditions:

- Baseline (no shock)
- 2008 Financial Crisis
- COVID Shock
- Sovereign Debt Crisis
- Oil Shock Stagflation
- Emerging Market Risk-Off Crisis
- Mild Recession
- Severe Recession
- Interest Rate Hike Cycle
- V-Shaped Recovery

For each scenario, the model generates a full distribution of predicted probabilities of default. The following summary statistics are computed:

- Mean PD
- Median PD
- 95th percentile PD
- Absolute change relative to baseline
- Relative change relative to baseline
- Mean absolute distribution shift

These metrics allow for both central tendency and tail-risk assessment under extreme but structured macroeconomic conditions, enabling a direct comparison of model behavior across different economic environments.

In addition to scenario-based stress testing, a systematic sensitivity analysis is performed by perturbing each macroeconomic variable individually. Each variable is shocked across a range from -3 to +3 standard deviations in increments of 0.5 standard deviations, while holding all other variables constant. For each perturbation level, the resulting change in mean predicted probability of default is recorded.

5. Results and Discussion

5.1 Data Exploration and Preparation

The raw Lending Club dataset exhibits only limited missingness, primarily concentrated in variables such as employment length and debt-to-income ratio. Overall, data quality is high, with no systematic gaps observed across key credit or borrower attributes.

The observed default rate is 15.2% overall; however, a marked distributional shift appears between training and test partitions. The training set contains a substantially higher

default rate (17.41%), while the test set drops to 3.57%. This divergence is driven by the time-based split strategy, where the test period corresponds to a more recent cohort of loans that have had less time to reach observed default status. This shift is an inherent consequence of temporal validation in credit risk settings and reflects realistic production constraints rather than a data quality issue.

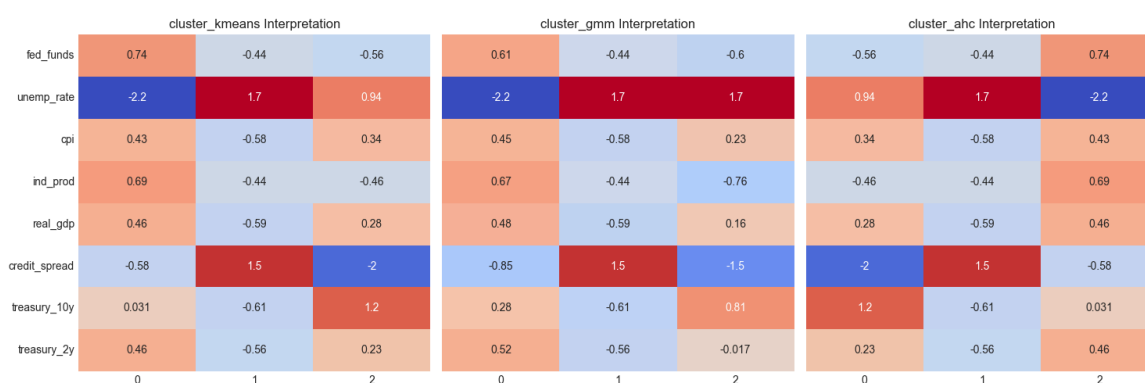
Following preprocessing and feature engineering, continuous variables exhibit improved statistical properties. Logarithmic transformations reduce right-skewness and bring heavy-tailed variables closer to normality, while standardization centers features around zero with consistent variance. Discretization of loan amounts into credit bands further reveals the underlying multimodal structure of lending tiers, improving interpretability and model stability.

After preprocessing, all missing values are fully resolved through median imputation, ensuring a complete and model-ready dataset without introducing structural bias.

5.2 Regime Detection Results

All three clustering methods (KMeans, GMM, and Agglomerative Hierarchical Clustering) consistently identify three macroeconomic regimes over the 2016-2018 period, with broadly aligned structure despite minor labeling differences.

The heatmap shows clear economic separation between regimes. One regime reflects expansion conditions (low unemployment, narrow credit spreads, and higher industrial production), another captures financial stress (high unemployment and wide credit spreads), and a third represents a transitional normalization phase between the two extremes.

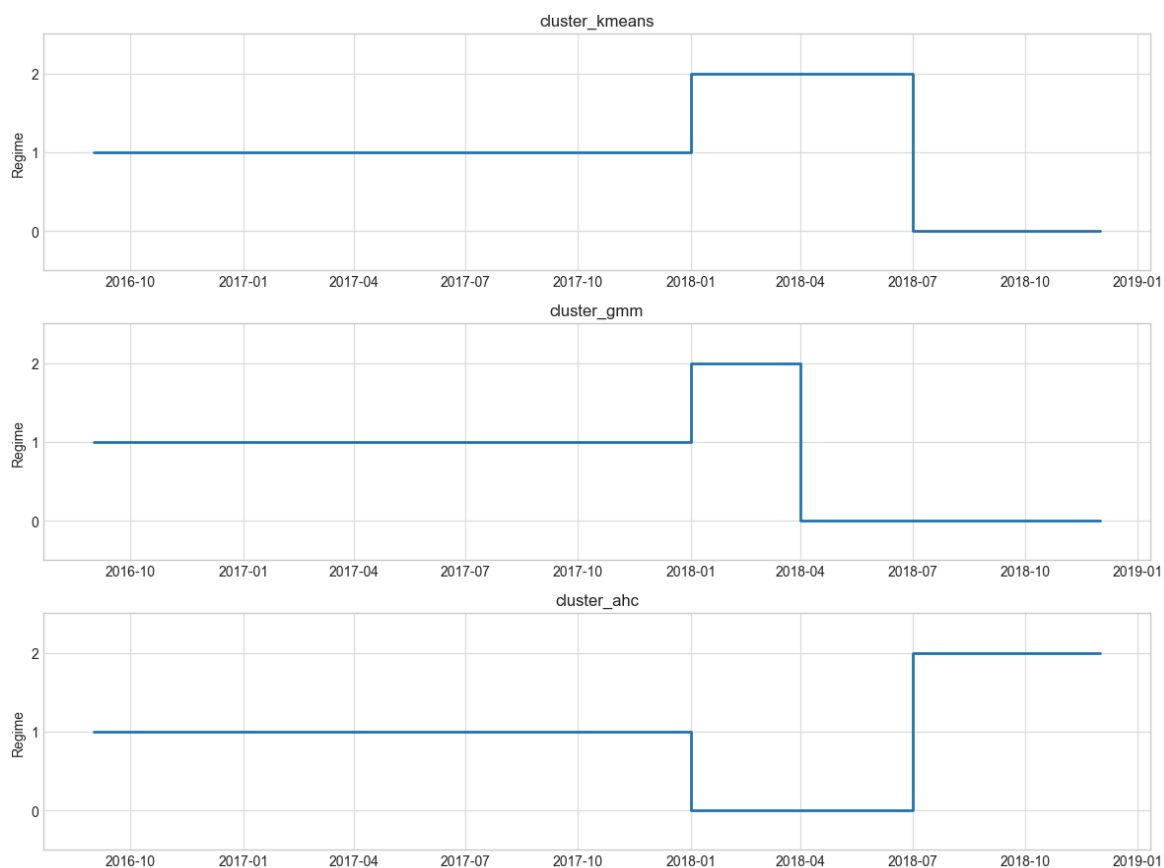


Heatmap of clusters

Unemployment rate is the strongest separating variable, followed by credit spreads, while CPI and Treasury yields provide weaker differentiation and act more as supporting signals.

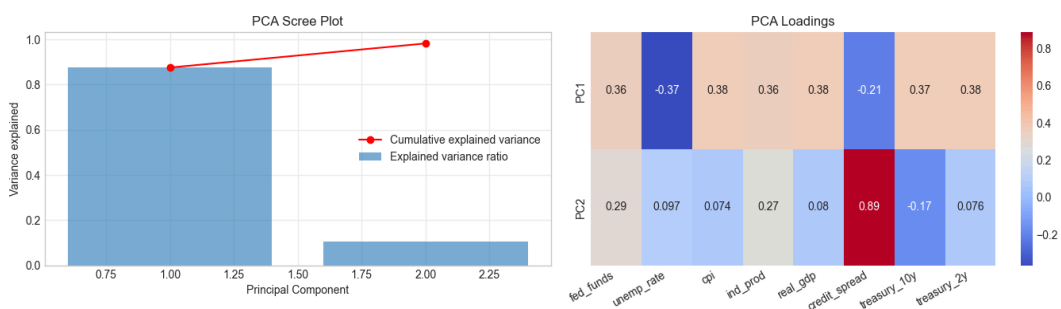
The timeline plots show a clear sequential pattern. KMeans and GMM produce nearly identical regime transitions: a low-rate regime dominates 2016-2017, a normalization phase appears in early 2018, and a tightening regime takes over in mid-to-late 2018.

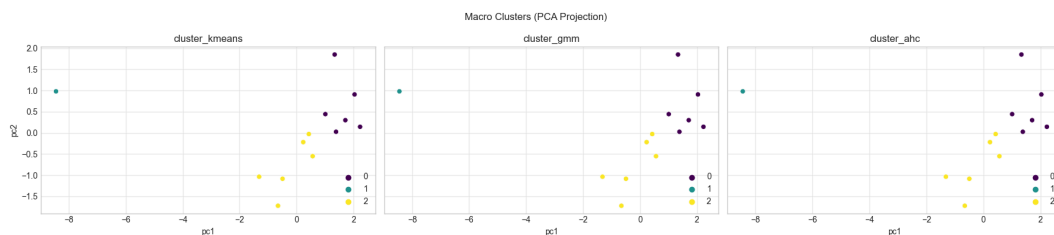
Hierarchical clustering follows the same structure but with slightly different timing boundaries.



Cluster detection in timeline

A majority-vote ensemble resolves these differences into a stable regime sequence that aligns with the Fed tightening cycle over the same period. Overall, the results indicate that macro conditions evolve in a simple three-phase structure, which also helps explain why raw macro variables provide limited additional predictive power beyond regime labels.





PCA Visualizations

PCA analysis confirms that the macroeconomic feature space is effectively near one-dimensional over the sample period. The first principal component (PC1) explains approximately 80% of total variance and functions as a broad “general conditions” factor, where most macro variables co-move in the same direction. The second principal component (PC2) accounts for around 15% of the variance and primarily captures a spread dynamic between credit risk measures and short-term interest rates.

This strong concentration of explanatory power in the first component indicates high multicollinearity among macro variables and explains the limited incremental value of raw macro features in predictive models. In practice, most of the macroeconomic signal is already embedded in a single latent time-driven factor, which reduces the independent informational content of the full variable set.

5.3 Framework and Model Performance

The table summarizes the key performance results across all model families and feature frameworks. XGBoost achieves the highest test ROC AUC in Framework 1 (0.707), indicating strong baseline discriminative performance using borrower-level features alone. However, when macroeconomic structure and regime information are incorporated, Logistic Regression in Framework 3 delivers the most balanced overall performance across ROC AUC, Average Precision, and Brier Score.

The starred entry highlights the recommended deployment specification, as it provides the best trade-off between predictive accuracy, probability calibration, interpretability, and robustness under temporal and macroeconomic shifts.

Model	Features Included	ROC-AUC	Avg Precision	Brier Score	Framework
Logistic Reg.	Borrower only	0.689	0.089	0.137	1 (Baseline)
Decision Tree	Borrower only	0.664	0.133	0.200	1 (Baseline)
XGBoost	Borrower only	0.707	0.207	0.068	1 (Baseline)
Random Forest	Borrower only	0.687	0.089	0.064	1 (Baseline)
Logistic Reg.	Borrower + Macro	0.704	0.062	0.108	2 (Macro)
XGBoost	Borrower + Macro	0.705	0.035	0.065	2 (Macro)
Logistic Reg.	Borrower + Macro + Regime	0.754	0.097	0.120	3 (Regime) ★
XGBoost	Borrower + Macro + Regime	0.703	0.037	0.065	3 (Regime)

Table: Model Performance Summary (threshold = 0.7)

Several consistent patterns emerge across all results.

First, incorporating raw macroeconomic variables (Framework 2) provides negligible improvement in ROC AUC relative to the borrower-only baseline, particularly for tree-based models. For Logistic Regression, gains are marginal. This suggests that macro variables, in their raw form, behave largely as a weak proxy for time rather than independent predictive signals. The limited sample window (2016-2018), combined with high collinearity among macro indicators, further reduces their incremental informational value.

Second, introducing a regime variable in Framework 3 materially improves Logistic Regression performance. ROC AUC increases to 0.754, representing gains of +0.050 over Framework 2 and +0.065 over Framework 1. At a decision threshold of 0.7, the model correctly identifies 33 true defaults, more than double the baseline specification. Calibration also improves in the mid-probability range (0.2-0.6), where decision uncertainty is highest in practice. This indicates that regime compression effectively transforms noisy macro signals into a structured, low-dimensional representation that is more usable for linear models.

Third, tree-based models remain largely invariant across frameworks. XGBoost shows near-identical performance across all specifications ($0.707 \rightarrow 0.705 \rightarrow 0.703$), and Random Forest follows the same pattern. Regime features do not appear among the most influential splits, indicating that these models already capture relevant nonlinear interactions internally and derive no additional benefit from explicit macro or regime augmentation.

Across all specifications, the most important borrower-level predictors remain interest rate, sub-grade, and grade, confirming that lender-assigned risk ratings dominate

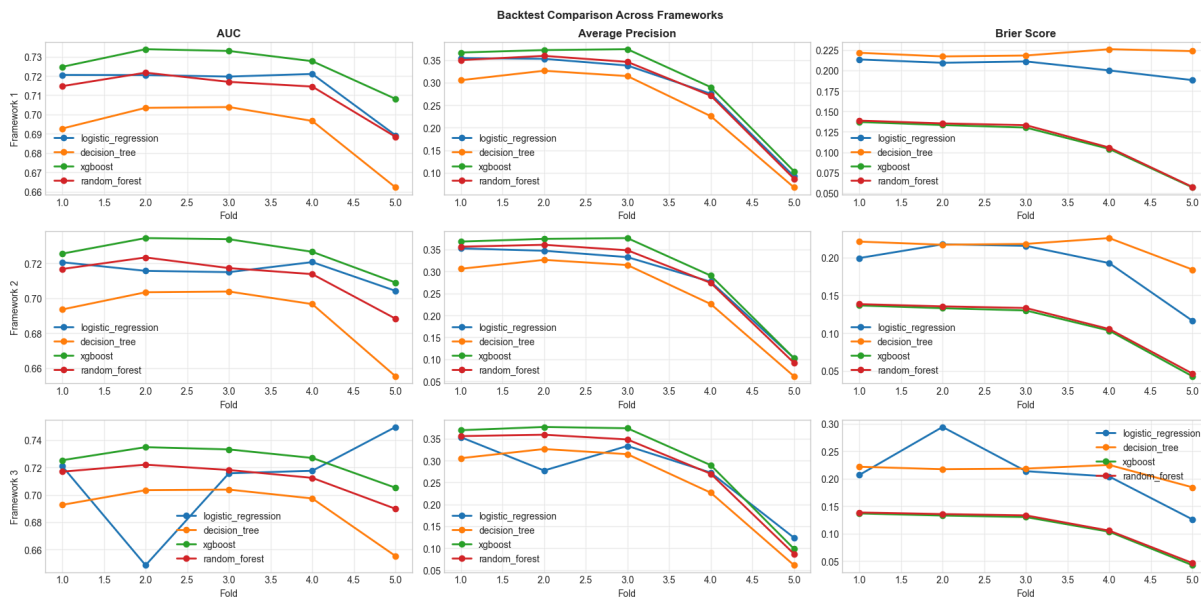
predictive signals. Within Framework 3, the regime variable emerges as one of the most influential features for Logistic Regression, alongside unemployment and inflation, reinforcing its role as a compact summary of macroeconomic conditions.

Overall, the results highlight a key structural distinction: raw macro variables are weak predictors in isolation, while regime-based transformations provide meaningful improvements for linear models by compressing correlated economic dynamics into a single informative factor.

5.4 Walk-Forward Backtesting Results

Walk-forward backtesting using five TimeSeriesSplit folds provides a more realistic evaluation of temporal stability. The results, summarized in Table 4, highlight consistent cross-framework patterns across all models and metrics.

A dominant feature across all specifications is a pronounced performance drop in Fold 5. This degradation is observed for every model, framework, and metric, indicating a systemic temporal shift rather than a model-specific failure. The most plausible explanation is a structural change in the data-generating process in the final period, potentially driven by evolving macroeconomic conditions in late 2018, loan maturity effects, or changes in underwriting standards.



Backtesting Results over Models and Frameworks

Framework 1 and Framework 2 behave almost identically across all models, reinforcing earlier findings that raw macroeconomic variables provide minimal contribution to temporal robustness. In contrast, Framework 3 improves Logistic Regression performance in the later folds, where regime shifts are more pronounced. This suggests that regime

encoding is most valuable precisely during periods of economic transition, when macro conditions are changing direction.

Tree-based models (Decision Tree, XGBoost, and Random Forest) remain largely insensitive to feature framework changes. XGBoost and Random Forest maintain stable performance across folds 1-4, particularly in calibration (Brier Score), but do not benefit from macro or regime augmentation. All models experience deterioration in Fold 5, confirming that the instability is driven by data shift rather than model design.

Overall, the backtesting results support a key conclusion: regime features enhance temporal robustness primarily for linear models, while ensemble methods already capture most stable structure internally.

Model	AUC Trend	Avg Precision	Brier Score	Framework Impact
Logistic Regression	Improves in F3, esp. late folds	Slight late-fold gain	Slight deterioration	F3 meaningfully improves LR
Decision Tree	No meaningful change	Stable, no gain	Minor improvement	Framework-insensitive
XGBoost	Near-identical across F1-F3	Strong drop in Fold 5	Very stable	No incremental value from F2/F3
Random Forest	Small early gain disappears	No consistent benefit	Slight calibration gain in F3	Converges across frameworks

Table: Walk-Forward Backtest Summary Across Models and Frameworks

5.5 Macro Stress Testing Results

The table below defines the 10 macroeconomic stress scenarios applied to the standardized input space. All scenarios are evaluated using the Framework 3 test set, which includes borrower-level features, macroeconomic variables, and the regime indicator.

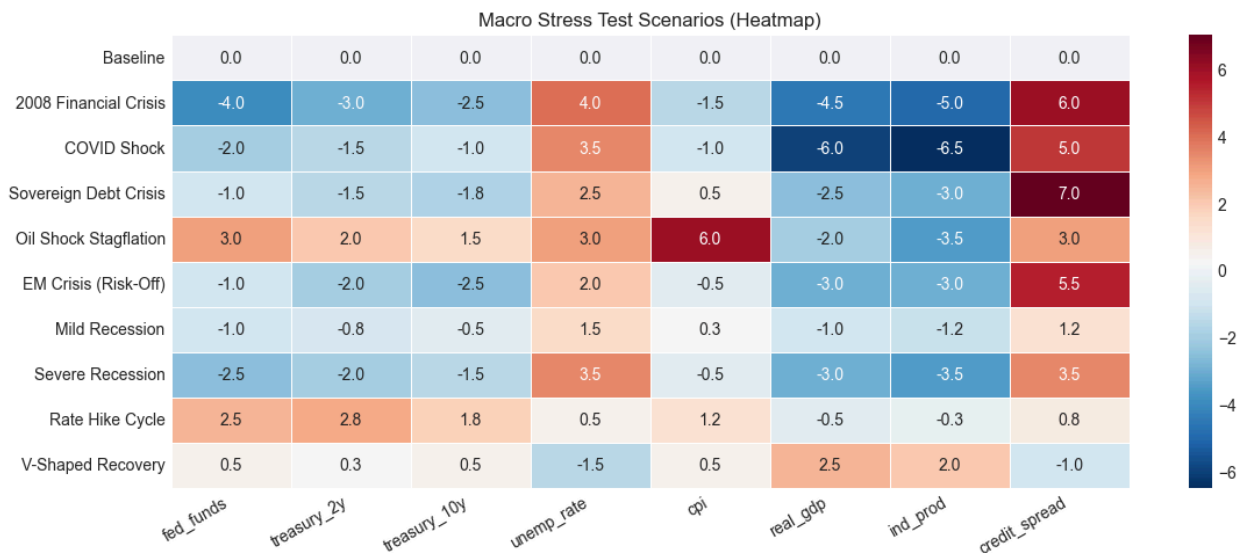
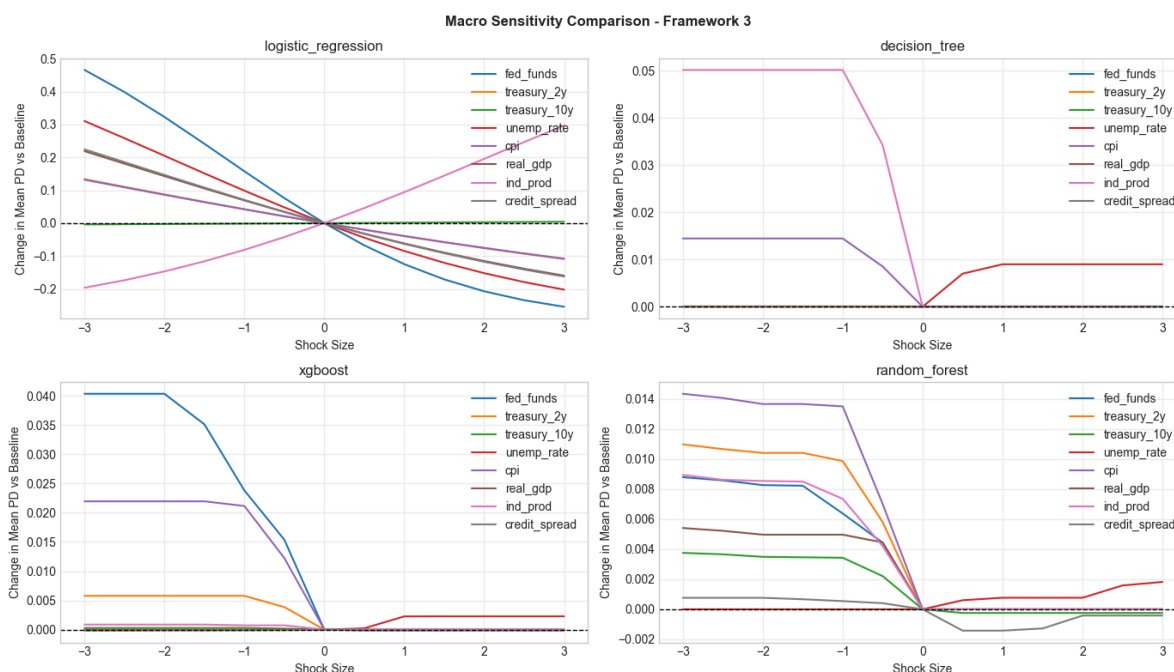


Table: Macro Stress Scenario Definitions

The most severe stress impact is observed under the 2008 Financial Crisis scenario, which produces the largest increase in probability of default across all macro-sensitive models. For Logistic Regression (Framework 3), mean PD rises from 0.13 under baseline conditions to 0.19, corresponding to an increase of approximately 46%, with the strongest effects concentrated in the upper tail of the distribution (95th percentile). The COVID Shock scenario generates a similar magnitude of stress response.

The Oil Shock Stagflation scenario exhibits a different transmission pattern, where elevated inflation and rising unemployment jointly drive a moderate increase in PD through both cost and demand channels. In contrast, the V-Shaped Recovery scenario leads to a reversion toward baseline PD levels, reflecting improved labor conditions and tighter credit spreads across all macro-sensitive specifications.

The Rate Hike Cycle scenario produces only a modest increase in PD despite significant interest rate shocks, suggesting that in this dataset higher rates are partially interpreted as a late-cycle expansion signal rather than a direct default trigger.



Model	Sensitivity Pattern	Primary Driver
Logistic Regression	Smooth, near-linear across all macro variables	fed_funds, treasury rates
Decision Tree	Threshold-driven, most variables inactive until split points	Sub-grade, int_rate dominate
XGBoost	Sparse, asymmetric - stronger to negative shocks	fed_funds (dominant)
Random Forest	Highly averaged, muted sensitivity across all variables	Short-term rates only

Macro Stress test sensitivities

Logistic Regression (Framework 3) shows smooth and economically consistent responses across all macro variables, with strongest sensitivity to the federal funds rate and Treasury yields. This aligns with standard credit risk intuition, where tighter monetary conditions increase borrower stress.

In contrast, tree-based models exhibit substantially weaker and less stable sensitivity patterns. XGBoost responds primarily to interest rate variables and predominantly under negative shocks, while Random Forest produces highly dampened responses across all macro dimensions. Decision Tree models show stepwise, threshold-driven behavior with limited activation across most shock ranges. As a result, while ensemble models perform well in predictive accuracy, they are less reliable for macro stress testing due to their weak and irregular sensitivity profiles.

5.6 Economic Interpretation and Implication

Our empirical findings in this paper support the hypothesis that credit default risk is state-dependent and is impacted by broader macro-economic conditions, and that borrower-only features do not fully explain default dynamics, especially during periods of increased systemic stress. Latent economic regimes impact default dynamics via multiple transmission channels, including labor market deterioration, tight credit conditions, declining borrower liquidity, and stringent refinancing constraints. As indicated by the results, during a stressful economic regime, even individuals with stable borrower characteristics may experience increased probabilities of default as a result of macroeconomic spillovers. The observed model performance improvements for the regime-aware model further suggest that non-linear economic dynamics are critical for accurate credit risk estimation and, therefore, the application of traditional static credit scoring models underestimates systemic default risk during periods of stressful macroeconomic conditions.

The findings in the paper demonstrate the value of incorporating machine learning frameworks with macro-financial regime detection techniques in credit risk modeling. The results give strong evidence that macroeconomic conditions and systemic regimes contain incremental predictive information, justifying the use of regime-sensitive machine learning frameworks in modern credit risk management. As such, financial institutions can benefit from incorporating latent economic state information into probability of default estimation frameworks to improve model robustness, calibration quality, and stress period forecasting performance.

6. Conclusion

This study developed and evaluated a macro-conditional and regime-dependent credit risk modeling framework using borrower-level data from the Lending Club dataset merged with macroeconomic indicators from the FRED database. A structured three-level feature hierarchy was constructed: borrower-only features, borrower features augmented with macroeconomic variables, and an extended specification incorporating latent economic regimes. These frameworks were assessed across four model families using walk-forward backtesting, macroeconomic stress testing, and sensitivity analysis.

The empirical results yield four main insights. First, raw macroeconomic variables provide limited standalone predictive gain within this dataset. Their contribution is largely diluted by low variation over the 2016-2018 sample window, causing them to behave similarly to a collinear temporal proxy rather than an independent source of predictive signal. Second, transforming macroeconomic data into discrete regime representations significantly improves model effectiveness in linear frameworks. In particular, Logistic Regression benefits substantially from regime conditioning, achieving an improvement of approximately 6.5 percentage points in ROC AUC relative to the borrower-only baseline.

Third, tree-based ensemble models exhibit comparatively limited sensitivity to additional macro or regime features. Their inherent ability to capture nonlinear interactions and implicit temporal structure reduces the marginal value of explicit macroeconomic augmentation. Fourth, all models exhibit a shared degradation in performance in the final temporal fold, indicating a structural regime shift in the underlying data rather than model-specific instability.

From an applied perspective, the most balanced solution is Logistic Regression with the regime-augmented feature set (Framework 3). This specification achieves strong predictive performance while maintaining calibration quality, interpretability, and stable behavior under macroeconomic stress scenarios. Unlike more complex ensemble methods, it also produces smooth and economically consistent response functions, making it particularly suitable for regulatory stress testing and point-in-time probability of default estimation.

Overall, the findings highlight that the effectiveness of macroeconomic information in credit risk modeling depends less on the raw variables themselves and more on how they are represented. Regime-based transformations convert noisy, high-dimensional macro signals into structured latent states that meaningfully improve model performance. This has direct implications for credit risk modeling under regulatory frameworks such as Basel, where both macro-conditioned stress testing and temporal consistency are essential requirements.

7. Future Direction

Several extensions can further enhance the robustness, interpretability, and generalizability of the proposed framework.

A natural extension is the use of a Hidden Markov Model (HMM) for regime detection over a longer macroeconomic time series. Unlike discrete clustering approaches, an HMM would produce probabilistic regime assignments rather than hard labels, allowing each observation to be associated with a distribution over latent states. This would enable soft regime conditioning in downstream credit risk models, where predicted default probabilities are weighted by regime likelihoods rather than a single deterministic state.

Another important direction is to evaluate the framework beyond consumer credit. Extending the analysis to corporate credit or SME lending would provide a stronger test of generalizability, particularly in settings where macroeconomic conditions exert a more direct and structural influence on default risk. Such an extension would also allow for richer heterogeneity in borrower sensitivity to economic cycles.

Finally, incorporating alternative and higher-frequency macro-financial data sources could significantly improve the informational content of the regime representation. Examples include yield curve transformations, option-implied volatility measures, and financial conditions indices. These variables provide forward-looking signals that are not fully captured by the current lagged FRED-based indicators, potentially improving early detection of regime shifts and enhancing model responsiveness to changing economic conditions.

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Appendix: GitHub and Code Repository

All project code is hosted publicly at:

<https://github.com/Taoufiq-Ouedraogo/Systematic-Credit-Risk-Modeling-under-Macroeconomic-Regimes>

The repository is organized as follows:

```
Systematic-Credit-Risk-Modeling-under-Macroeconomic-Regimes/
├── modules/
│   ├── dataprep.py
│   │   (Contains full data engineering pipeline)
│   ├── models.py
│   │   (Implements core modeling workflow)
│   ├── backtest.py
│   │   (Provides time-series backtesting framework based on walk-forward validation)
│   ├── stresstest.py
│   │   (Implements macroeconomic stress testing and sensitivity analysis procedures)
│   └── visualisations.py
│       (Contains functions for exploratory data analysis and model diagnostics)
└── notebook/
    └── Systematic-Credit-Risk-Modeling-under-Macroeconomic-Regimes.ipynb
        (Serves as the end-to-end orchestration layer)
```